ABHISHEK RANGI

github.com/Abhi-rangi   linkedin.com/in/abhishek-rangi   Portfolio

abhishekrangi09@gmail.com   +1(609)-255-7119

EDUCATION

Master of Science, Computer Science — Aug 2023 - May 2025
Towson University, Towson, MD — CGPA - 3.96/4.0
Advisor: Dr. Yeong-Tae Song
Thesis: On-device AI on medical data

Bachelor of Technology, Computer Science & Engineering — July 2018 - July 2022
Keshav Memorial Institute of Technology, Hyderabad (Affiliated to JNTUH, India) — CGPA - 7.52/10

SKILLS

Programming Languages	Python, C/C++, Java
AI/ML	LLMs, LangChain, RAG, TensorFlow, scikit-learn, OpenCV, Data Processing
Database	SQL, SQL Server, SQLite, MongoDB, PostgreSQL
Web Technologies	TypeScript, AngularJS, React, NextJS, Node JS, jQuery
Tools	GitHub, Bitbucket, JIRA, Jenkins, Google Suite, Microsoft Office Suite, Docker

WORK EXPERIENCE

Graduate Research Assistant
Towson University, Bare Machine Computing (BMC) Research — Aug 2023 – May 2025
- Assisted in developing a chat application utilizing C, C++ and assembly language, contributing to complex software solution engineering.
- Surveyed over 100 research publications to identify cybersecurity vulnerabilities and attacks in machines with operating systems, comparing these findings with Bare Machine Computing to highlight its security benefits.
- Supervised students during semesters, enhancing course delivery and student engagement through improved educational techniques and interaction strategies.

Full Stack Developer
DBS Tech India, Consumer Banking & Core Engines Technology (C2E) — Aug 2022 – July 2023
- Developed scalable RESTful services using Java and integrated SQL/NoSQL databases including PostgreSQL, improving system responsiveness across distributed services.
- Built interactive Single Page Applications (SPAs) using AngularJS and TypeScript, incorporating jQuery and Bootstrap for enhanced UX.
- Designed and implemented reusable microservices for banking operations, deploying them via Jenkins CI/CD pipelines to Kubernetes clusters.
- Collaborated with QA and support teams to triage and resolve customer-reported issues, performing root cause analysis and providing Tier 2/3 support.
- Conducted rigorous unit and integration testing using Jasmine and JUnit to ensure consistent and reliable deployments.

Blockchain Developer Intern
Virtusa, Technology — July 2021 – April 2022
- Built and debugged over 13 backend APIs in Node.js (Express), including authentication, data processing, and blockchain interaction endpoints.
- Managed and queried large-scale MongoDB datasets to support secure blockchain-based supply chain workflows.
- Participated in Agile sprints and documented technical issues and resolutions, assisting the support team in cross-functional bug tracking and resolution.
- Integrated APIs with frontend interfaces using REST standards, improving full-stack synchronization and system robustness.

RESEARCH & PUBLICATIONS

- **On-Device AI for Secure Patient Health Monitoring**. Abhishek Rangi, Yeong-Tae Song. *23rd IEEE/ACIS International Conference on Software Engineering, Management and Applications (SERA)*, 2025.
- **An Evaluation of the Security of Bare Machine Computing (BMC) Systems Against Cybersecurity Attacks**. Fahad Alotaibi, Ramesh K. Karne, Alexander L. Wijesinha, Nirmala Soundararajan, **Abhishek Rangi**. *Journal of Cybersecurity and Privacy*, 4(3):678–730, 2024. [Paper]

- **A Chat Application on a Bare Internet**. Fahad Alotaibi, Ramesh K. Karne, Alexander L. Wijesinha, Nirmala Soundararajan, **Abhishek Rangi**. In *Proceedings of the 2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)*, 2024. [Paper]

PROJECTS

Smart Library Assistant

- Identified limitations with traditional fine-tuning methods when developing a library chatbot and pivoted to Retrieval-Augmented Generation (RAG), ensuring more accurate, context-aware, and personalized responses.
- Integrated FAISS vector storage for efficient semantic retrieval and leveraged HuggingFaceEmbeddings to significantly boost document matching accuracy, enhancing user satisfaction.
- Optimized chatbot responsiveness by implementing Redis for caching, while orchestrating conversational flows and dynamic deployment using Chainlit, improving interaction quality and scalability.
- Built a robust ConversationalRetrievalChain architecture with a large language model, ensuring seamless context maintenance across user interactions and delivering a highly adaptive user experience.

Analysis of 911 Calls for Detroit and New York City

- Launched a data-driven project analyzing emergency 911 calls from two major cities, aiming to predict hotspots and improve urban public safety operations.
- Applied advanced modeling techniques such as SARIMAX for time-series forecasting, DBSCAN for anomaly detection, and K-Means clustering to uncover spatial and temporal incident patterns.
- Devised a methodology combining predictive modeling and spatial analysis, generating insights on call frequency and distribution to inform strategic emergency resource allocation.
- Delivered actionable findings expected to aid public agencies in implementing targeted interventions, ultimately supporting safer urban environments and optimized emergency response systems.

Inventory Management System

- Designed and developed an advanced inventory management solution for a simulated retail business ("Bharat Ki Saree"), aimed at improving operational visibility and stock control.
- Built a relational database using SQL and Node.js, integrating real-time inventory tracking, client management, and automated ordering features tailored for business growth.
- Automated key processes through SQL triggers, ensuring immediate and error-free updates for inventory changes and order fulfillment, thus reducing operational overhead.
- Created a user-friendly Angular interface for administrative users, enhancing the system's usability and enabling effective inventory and order management with minimal technical training.

Secure Cloud Storage using Cryptography

- Addressed critical cloud security challenges by designing a secure file storage system integrating Flask, AWS S3, and hybrid cryptographic techniques for high-stakes environments like military and industry.
- Implemented a hybrid encryption mechanism combining AES, DES, RSA, along with steganography, ensuring robust multi-layered protection for sensitive data.
- Devised a file-slicing encryption strategy to maximize throughput while minimizing encryption time, while LSB steganography was used for secure key management.
- Enabled highly secure cloud file storage and retrieval via AWS Lambda automation, boosting system reliability and strengthening user trust in cloud-based data solutions.

LEADERSHIP & ACTIVITIES

Lab Administrator, CIS TECH HUB, Towson University — Jan 2024 - Present
Member, TU Software Engineering Club, Towson University — Aug 2023 - Present
Member, International Students Association, Towson University — Aug 2023 - Present